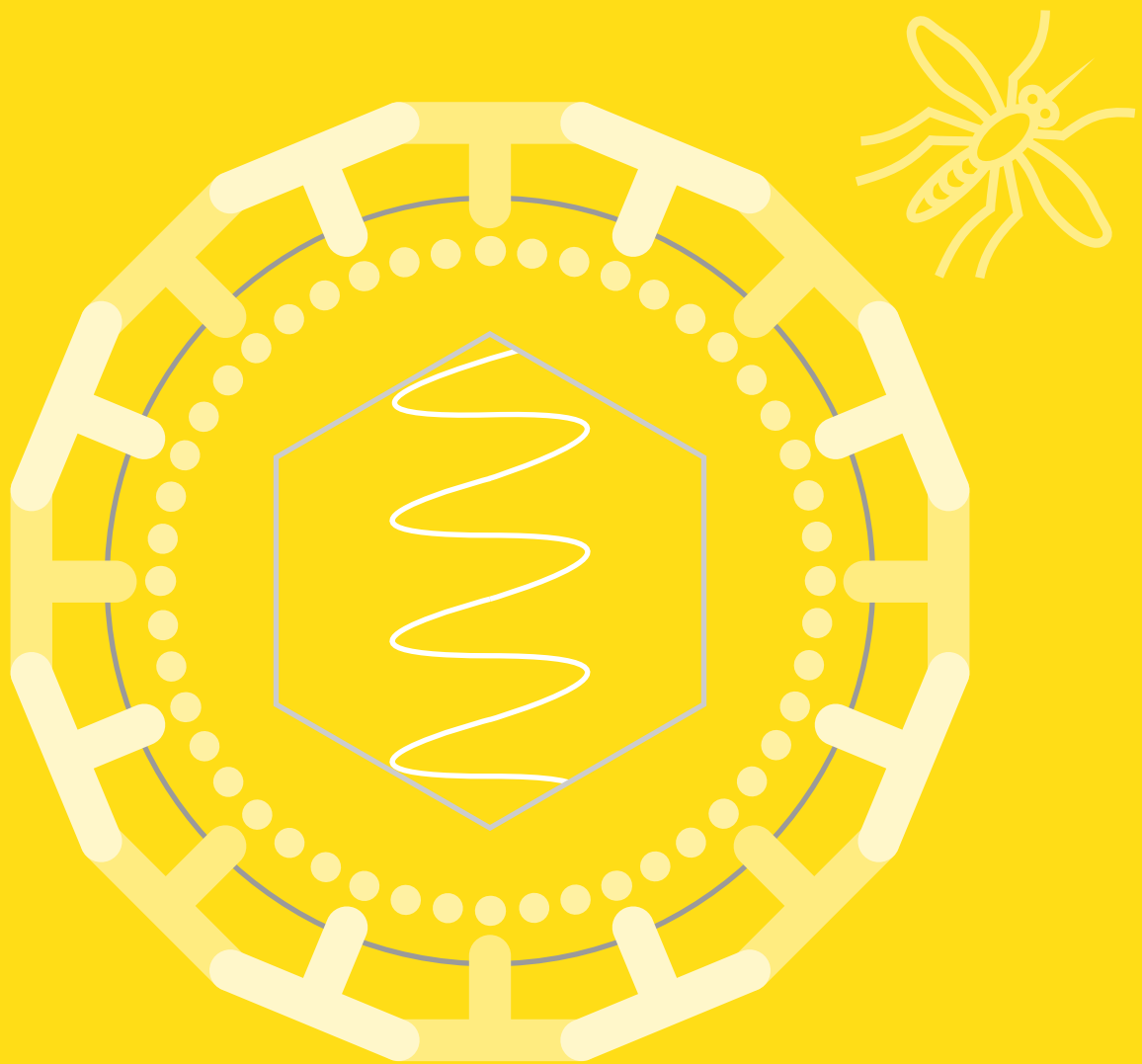


Root cause analysis of low yellow fever vaccination coverage

User manual and questionnaire



Root cause analysis of low yellow fever vaccination coverage

User manual and questionnaire

Root cause analysis of low yellow fever vaccination coverage: user manual and questionnaire

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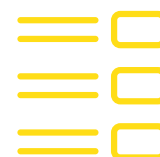
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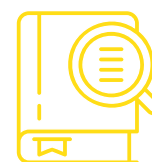
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WHO gratefully acknowledges all WHO staff, partners, and country teams whose efforts contributed to the development of this tool.

Abbreviations and acronyms



CDC	United States Centers for Disease Control and Prevention
EPI	Essential Programme on Immunization
TDR	UNICEF/UNDP/World Bank/WHO Special Programme for Research and Training on Tropical Diseases
UNDP	United Nations Development Programme
UNICEF	United Nations International Children's Fund
WHO	World Health Organization



Glossary of terms

Community: a group of people connected by common characteristics, such as geographic location, age, gender, profession, ethnicity, faith, shared vulnerability or risk, or shared interests and values (1).

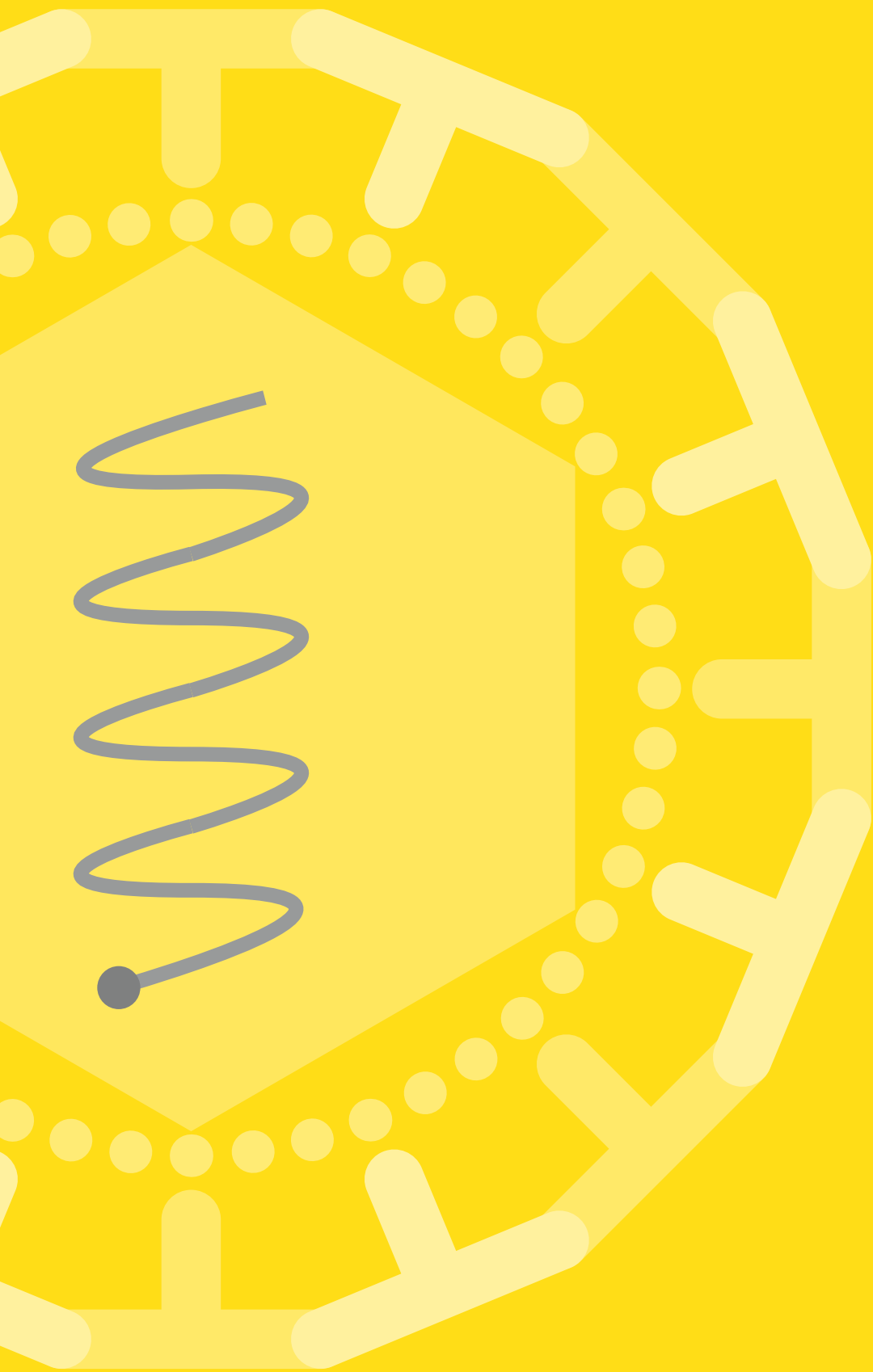
Preventive mass vaccination campaign: a strategy to rapidly raise population immunity and reduce the number of susceptible individuals to achieve disease control or elimination goals.

Stakeholders: governments and community leaders that have a vested interest in protecting the health of their own country, region or community (1).

Vaccine hesitancy: delay in acceptance or refusal of safe vaccine despite availability of vaccination (2).

Reactive vaccination campaign: a strategy implemented in response to an outbreak of a vaccine-preventable disease to rapidly raise immunity in a specific population or area where the disease has been detected and to control its spread.

Yellow fever immunity gap: a reduction in population-wide immunity to yellow fever. This can occur when a significant portion of the population has not been exposed to the yellow fever virus, either through natural infection or vaccination, leading to low herd immunity. The immunity gap can be estimated with the aid of the immunity gap analysis tool (WHO, unpublished).





Introduction

Despite significant progress in reducing yellow fever outbreaks through the Eliminate Yellow fever Epidemics strategy, inadequate vaccination coverage in many countries is creating a growing immunity gap. This threatens past investments in yellow fever control and increases the risk of outbreaks, potentially leading to wider disease spread.

While 80% immunization coverage is needed for herd immunity (3), many at-risk countries have routine immunization rates below 50%. This has fueled a resurgence of yellow fever in Africa, with 29 outbreaks reported in 15 countries since 2020 (4,5).

To address this urgent threat, targeted vaccination campaigns are planned for adolescents and adults in high-risk countries where routine immunization and past campaigns have failed to achieve sufficient coverage. These campaigns will complement, not replace, efforts to strengthen routine immunization.

Prior to these campaigns, a detailed analysis of immunity gaps and their root causes will be conducted in high-risk areas. This assessment will provide crucial information for country-level planning and applications for support. Ultimately, the findings will be used to develop strategies to reinforce routine immunization, benefiting the delivery of all vaccines.



1. Purpose of the tool

The tool aims at identifying the root causes of low yellow fever vaccination coverage in a district with a high immunity gap ahead of a planned targeted yellow fever vaccination campaign. Low yellow fever vaccination coverage may be related to impaired implementation and/or reporting of routine immunization and/or mass campaigns if any were conducted in the district. At present, two language versions of the tool have been developed (English and French).

2. Methodology development

Literature review

We conducted a scoping review of quantitative and qualitative studies reporting on yellow fever burden and epidemiology, vaccination, or factors leading to yellow fever transmission and outbreak. Our database search included PubMed, that combined subject heading searches with free-text keyword searches. We did not restrict our review to a specific publication period or geographic location of the outbreaks. The included studies were published in English and French, or in other languages with English abstracts. Additionally, we reviewed the reference lists of relevant studies to identify further research. This review included both published and unpublished articles. We also reviewed World Health Organization (WHO) guidelines on yellow fever, internal tools, weekly epidemiological reports, WHO arboviruses and immunization guidelines, national investigation reports, and instructions from each WHO pre-qualified yellow fever vaccine manufacturer. For the root cause analysis method, we reviewed internal documents and root cause analysis guidelines. The list of published documents reviewed is provided in Annex 1.

Pilot testing

A first version of the tool was piloted in three yellow fever high risk countries fulfilling two criteria: history of preventive mass vaccination campaign and resurgence of yellow fever cases requiring a reactive vaccination campaign in 2022¹. The three countries were: Guinea (September-October 2023); Cameroon (November-December 2023); and the Central African Republic (March-April 2024).

Peer review

A second version of the tool was peer-reviewed after the pilot testing phase by experts from the Eliminate Yellow fever Epidemics vaccine delivery working group and risk analysis working group, the WHO department of Immunization, Vaccines and Biologicals, the United States Centers for Disease Control and Prevention (CDC) and the Africa Centers for Disease Control and Prevention.

All external experts who reviewed the tool submitted a WHO declaration of interest form, disclosing potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence concerning the subject.

3. Target audience

The tool will be useful to all national experts working at the Essential Programme on Immunization (EPI) to identify priorities for strengthening capacity in districts with low yellow fever vaccination coverage. It will also allow WHO country and regional offices to guide technical support to countries and to identify priorities for strengthening capacity.

¹ For the current yellow fever confirmed case definition, please refer to the WHO yellow fever vaccine-preventable diseases surveillance standard manual (2020) (6).

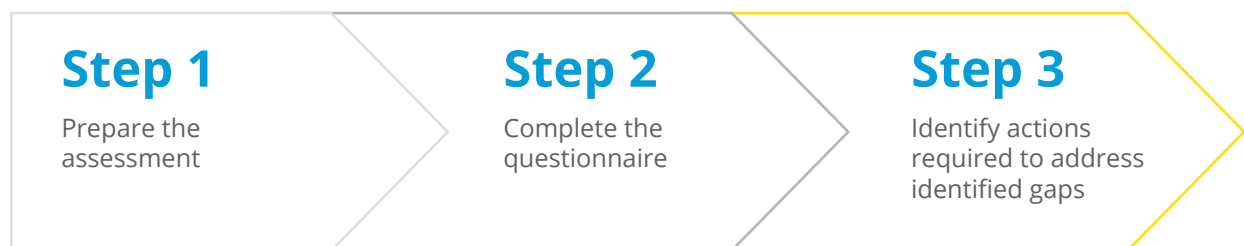
4. When to use the tool?

The tool should be deployed by (1) countries planning to implement a yellow fever targeted vaccination campaign following an immunity gap analysis, and (2) in a district with a high immunity gap that plans to conduct a yellow fever targeted vaccination campaign. This district is identified with the aid of the immunity gap analysis tool (WHO, unpublished). The findings of this analysis will help the district to develop a recovery plan to strengthen yellow fever routine immunization.

5. How to use the tool?

Fig. 1 illustrates the steps to conduct the root cause analysis of low yellow fever vaccination coverage in a district with a high immunity gap.

Fig. 1. Three-step procedure to conduct the root cause analysis of low yellow fever vaccination coverage.



Step 1. Prepare the assessment

Form a team

Evaluation should be led by EPI staff at regional or national level with support from the WHO/United Nations Children's Fund (UNICEF) country office. The team that conducts the root cause analysis should include health personnel from the peripheral and central levels. This is essential as information is required from all levels and also because they are responsible for implementing corrective actions. It is important for representatives of the community and authorities to be involved as they make the team's work easier.

The team should be composed of staff from the national EPI, staff from regional and district health levels involved in the supervision of routine immunization, and staff from the WHO and UNICEF country offices involved in vaccination programmes. The use of this tool does not require specific additional training. All team members must provide proof that they have been vaccinated against yellow fever at least 10 days before the start of the field investigation.

Identify the district and the health centre to be assessed

The district in which the tool is used is selected with the aid of the immunity gap analysis tool (WHO, unpublished). This represents the district with a high immunity gap where the national team intends to conduct a targeted vaccination campaign. **Note:** If several districts present a high immunity gap, the team refers to the immunity gap analysis tool to help to prioritize the district to be investigated.

The health centre under evaluation includes active participation in vaccination activities and is part of the health area with the lowest yellow fever vaccination coverage in the district under investigation.

Estimate the required budget for the field assessment

In the preparatory stage, it is essential to accurately forecast the budget for the field assessment. Costs that need consideration include expenses for transport and fuel, expenses for data collection materials (such as pencils and questionnaire photocopies), and the fees for those conducting the assessment. Additional costs may be considered, subject to the context. A template to help budget cost planning is provided in Annex 2.

Step 2. Complete the questionnaire

The questionnaire is divided into the following two key sections.

Section A (yellow fever vaccination coverage estimates in the district) is designed to report the estimated yellow fever vaccination coverage of routine activities at central, district and health centre's levels. It is also designed to determine whether yellow fever mass vaccination campaigns have been implemented in the district under investigation and, if confirmed, to report the estimated vaccination coverage.

Section B (evaluation of the root causes of low vaccination coverage) is designed to investigate the chain of causality of the low yellow fever vaccination coverage in the district under investigation leading to a high immunity gap. The chain of causality is determined using a phased approach, consistent with the approach detailed in the WHO measles outbreak root cause analysis guide 2021 (7).

Failure to vaccinate contributing to the low yellow fever vaccination coverage are categorized as policy-based reasons, provider-based reasons and user-based reasons (Table 1). Investigation of these three categories is undertaken into second-level causes, then lower-level causes through a series of 'why' questions, where the answer to each 'why' question leads to another 'why' question that reveals the chain of causality down to the most fundamental causes.



Table 1. Immediate and underlying causes related to low yellow fever vaccination coverage in a district.

Failure to vaccinate		
Reasons	Immediate causes	Underlying causes
Policy-based	• Inaccuracy of estimation of yellow fever vaccination coverage • Gap in the vaccination plan for general populations and population at higher risk for yellow fever	
Provider-based	Routine activities	• Missed routine vaccination sessions • Stock-out of yellow fever vaccines/vaccination equipment • Lack of qualified vaccinators • Lack of knowledge/misconceptions regarding: – the yellow fever vaccination policy – the coadministration of the yellow fever vaccine with another antigen – the contraindications of the yellow fever vaccine • Other reasons for not being vaccinated against yellow fever • Limitation in skills enhancement • Lack of supervision of vaccination activities
	Mass vaccination campaigns	• Lack of knowledge/misconceptions regarding: – the coadministration of the yellow fever vaccine with another antigen – the contraindications of the yellow fever vaccine • Other reasons for not being vaccinating against yellow fever: – stock-out of vaccines – district not fully covered by the vaccination campaign • Limitation in skills enhancement • Lack of supervision
User-based	• Vaccines hesitancy/obstacles to vaccination activities (routine and campaigns) • yellow fever vaccines knowledge and misconceptions • obstacles to routine vaccination activities • lack of awareness of the mass vaccination campaigns	

Sections are clickable for easier navigation



Data are collected: (1) in the district under investigation, and (2) in one of the district health centres. Criteria include active participation in vaccination activities and currently covering the health area with the lowest vaccination coverage.

Data are gathered through a document review, one-on-one conversations and group discussions. A document review should include: (1) documents shared by WHO; (2) documents shared by the country; and (3) documents available at district and health facility levels. One-on-one conversations are conducted with staff working in vaccination activities. The position of the staff involved in the vaccination activities and interviewed by the data collection team is mentioned within the questionnaire prior to each question. Finally, group discussions are conducted in the village where most confirmed yellow fever cases live.²

A list of sources to find information that facilitates the recording of data is provided at the beginning of each section of the part B questionnaire under the section “data sources” as shown below in Box 1.

Box 1. Example of data sources for finding information.

SECTION B. Evaluation of the root causes of low yellow fever vaccination coverage

B.1. Policy-based reasons

Purpose: These questions assess the specific health policies that may have led to the failure to vaccinate with the yellow fever vaccine.

B.1.1 Inaccuracy of estimation of yellow fever vaccination coverage

Data sources for finding information:



Vaccination procedure regulations (current immunization protocol standard, current national vaccination strategic plan)



Vaccination records at the health centre level (health centre offering vaccination sessions and belonging to the health area with the lowest yellow fever vaccination coverage)



Interviews with the vaccinators working in the health centre offering vaccination sessions and belonging to the health area with the lowest yellow fever vaccination coverage

The procedures used for data collection are indicated within the questionnaire prior to each question. The investigators proceed to the questionnaire with previous mentions of DR (document review), EPI C (interview with the EPI focal point at the central level), EPI D (interview with the EPI focal point at district level), EPI HC (interview with the EPI focal point at the health centre level), Dir (interview with the director of the health centre), Log C (interview with the logistician of the vaccine storage room at the central level), Log D (interview with the logistician of the vaccine storage room at the district level), Log HC (interview with the logistician of the vaccine storage room at the health centre level), CHW (interview with community health workers), and GD (group discussions).

² For the current yellow fever confirmed case definition, please refer to the WHO yellow fever vaccine-preventable diseases surveillance standard manual (2020) (6).

Step 3. Identify actions required to address identified gaps

The final step involves identifying actions that address the gaps identified by the root cause analysis. This feeds into the recovery plan to strengthen yellow fever routine immunization in the district.

6. Timeline

The phase of data collection is completed over a period of 5 days maximum (travel to the district and health centre included). The data analysis phase is completed over 2 days.

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Annex 1.

List of documents included in the literature review

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Annex 2.

Budget planning for the investigation

Tables A2.1 to A2.4 are intended to help plan a budget for the evaluation of root causes of low yellow fever vaccination coverage.

Table A2.1. Staffing budget.

Personnel	Number	Number of days	Total	Rate (country currency)	
				Per day	Total
EPI focal point at central level					
EPI focal point at district level					
WHO country office National Professional Officer					
Other personnel					
Subtotal					

Table A2.2. Supplies budget

Items	Price/units (country currency)	Number of units	Total
Pencils			
Photocopies			
Other supplies			
Subtotal			

Table A2.3. Transport budget

Items	Price/units (country currency)	Number of units	Total
Vehicle			
Fuel			
Subtotal			

Table A2.4. Communication.

Items	Price/units (country currency)	Number of units	Total
Telephone			
Other communication items			
Subtotal			

Evaluation total budget (country currency): _____



Annex 3.

Evaluation of root causes of low yellow fever vaccination coverage -Questionnaire -

Abbreviations and acronyms

EPI	Essential Programme on Immunization
MM/YYYY	month/year
PMVC	preventive mass vaccination campaign
RVC	reactive vaccination campaign
TDR	UNICEF/UNDP/World Bank/WHO Special Programme for Research and Training on Tropical Diseases
UNDP	United Nations Development Programme
UNICEF	United Nations International Children's Fund
WHO	World Health Organization

Glossary of terms

Community: a group of people connected by common characteristics, such as geographic location, age, gender, profession, ethnicity, faith, shared vulnerability or risk, or shared interests and values (1).

Preventive mass vaccination campaign: a strategy to rapidly raise population immunity and reduce the number of susceptible individuals to achieve disease control or elimination goals.

Stakeholders: governments and community leaders that have a vested interest in protecting the health of their own country, region or community (1).

Vaccine hesitancy: delay in acceptance or refusal of safe vaccine despite availability of vaccination (2).

Reactive vaccination campaign: a strategy implemented in response to an outbreak of a vaccine-preventable disease to rapidly raise immunity in a specific population or area where the disease has been detected and to control its spread.

Yellow fever immunity gap: a reduction in population-wide immunity to yellow fever. This can occur when a significant portion of the population has not been exposed to the yellow fever virus, either through natural infection or vaccination, leading to low herd immunity. The immunity gap can be estimated with the aid of the immunity gap analysis tool (WHO, unpublished).

Questionnaire outline

The tool is divided into the following two key sections.

Section A (yellow fever vaccination coverage estimates in the district) is designed to report the estimated yellow fever vaccination coverage of the routine activities at central, district and health centre levels. It is also designed to determine whether yellow fever mass vaccination campaigns have been implemented in the district under investigation and, if confirmed, to report the estimated vaccination coverage.

Section B (evaluation of the root causes of low yellow fever vaccination coverage) is designed to investigate the chain of causality of low yellow fever vaccination coverage in the district under investigation leading to a high immunity gap. The chain of causality is determined using a phased approach, consistent with the approach detailed in the World Health Organization (WHO) measles outbreak root cause analysis guide published in 2022 (3).

Section B includes three major categories that investigate the reasons for low yellow fever coverage: (1) vaccination policy; (2) provider; and (3) user. Investigations of these three categories delve into the second-level causes, followed by lower-level causes through a series of 'why' questions, where the answer to each 'why' question leads to another 'why' question that reveals the chain of causality down to the most fundamental causes.

The immediate and underlying probable causes related to a low vaccination coverage are listed in Table. 1.

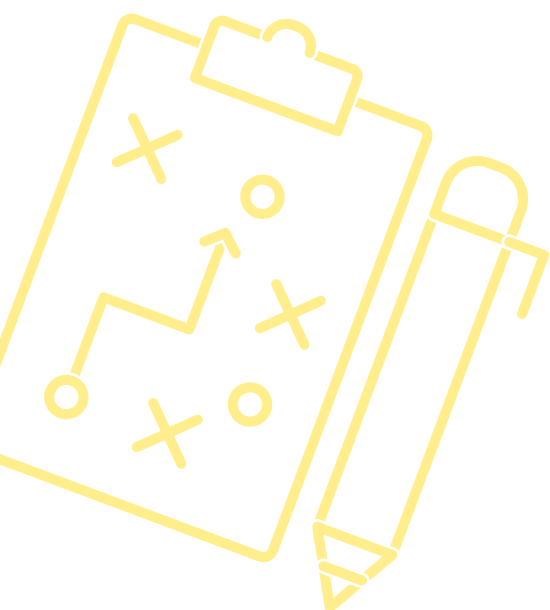


Table 1. Immediate and underlying causes related to low yellow fever vaccination coverage in a district.

Failure to vaccinate		
Reasons	Immediate causes	Underlying causes
Policy-based	• Inaccuracy of estimation of yellow fever vaccination coverage • Gap in the vaccination plan for general populations and population at higher risk for yellow fever	
Provider-based	Routine activities • Missed routine vaccination sessions • Stock-out of yellow fever vaccines/vaccination equipment • Lack of qualified vaccinators • Lack of knowledge/misconceptions regarding: – the yellow fever vaccination policy – the coadministration of the yellow fever vaccine with another antigen – the contraindications of the yellow fever vaccine • Other reasons for not being vaccinated against yellow fever	• Limitation in skills enhancement • Lack of supervision of vaccination activities
	Mass vaccination campaigns • Lack of knowledge/misconceptions regarding: – the coadministration of the yellow fever vaccine with another antigen – the contraindications of the yellow fever vaccine • Other reasons for not being vaccinating against yellow fever: – stock-out of vaccines – district not fully covered by the vaccination campaign	• Limitation in skills enhancement • Lack of supervision
User-based	• Vaccines hesitancy/obstacles to vaccination activities (routine and campaigns) • yellow fever vaccines knowledge and misconceptions • obstacles to routine vaccination activities • lack of awareness of the mass vaccination campaigns	

Sections are clickable for easier navigation



The three major categories that explore a plausible underlying cause for a low yellow fever vaccination coverage include the data source for finding information and the information to record to allow a subsequent analysis (Box 1).

Box 1. Example of data sources for finding information.

SECTION B. Evaluation of the root causes of low yellow fever vaccination coverage

B.1. Policy-based reasons

Purpose: These questions assess the specific health policies that may have led to the failure to vaccinate with the yellow fever vaccine.

B.1.1 Inaccuracy of estimation of yellow fever vaccination coverage

Data sources for finding information:



Vaccination procedure regulations (current immunization protocol standard, current national vaccination strategic plan)



Vaccination records at the health centre level (health centre offering vaccination sessions and belonging to the health area with the lowest yellow fever vaccination coverage)



Interviews with the vaccinators working in the health centre offering vaccination sessions and belonging to the health area with the lowest yellow fever vaccination coverage

The procedures used for data collection are indicated in the questionnaire prior to each query within the tool. Box 2 lists the abbreviations of the procedures.

Box 2. Abbreviations of the procedures for data collection.

DR: document review

EPI D: interview with the EPI (Essential Programme on Immunization) focal point/supervisor at district level

EPI HC: interview with vaccinators at the health centre level

Dir: interview with the director of the health centre

Log C: interview with the logistician of the vaccine storage room at the central level

Log D: logistics maintenance of the vaccine storage room at the district level

Log HC: logistician maintenance of the vaccine storage room at the health centre

CHW: community health workers

GD: group discussions

To evaluate the root causes for low yellow fever vaccination coverage in a district with a high immunity gap, data are collected:

- In the district under investigation and
- In one of the health centres within that district, which actively participates in vaccination activities and currently covers the health area with the lowest yellow fever vaccination coverage.

Data are gathered through either a document review, one-on-one conversations or group discussions.

Procedures to conduct the document review

Instructions for conducting the document review are provided at each stage of the tool where the review of documents is necessary.

Procedures to conduct the one-on-one conversations

The semi-structured questionnaires are conducted with vaccination staff involved in vaccination activities. The questionnaires are designed with closed and open-ended questions, allowing flexibility in the interview. The interviewer can probe and follow different directions as information emerges, including flexibility in the sequence of questions. If a response is unclear, the interviewer asks for clarification to ensure to understand the participant's perspective. It is important to minimize notetaking and to consider a voice recorder (with the participant's permission) to capture the conversations.

Procedures to conduct the group discussion

The group should represent caregivers of children who are of age to be vaccinated against yellow fever through the activities of the Expanded Programme on Immunization (EPI) in order to understand why children are not vaccinated against yellow fever. The group should comprise 8-10 caregivers. Ideally, caregivers of three types of children should be included:

- children that were never vaccinated with pentavalent 1;
- children that received pentavalent 1, but did not receive yellow fever vaccine before the evaluation;
- children that did receive yellow fever vaccine before the evaluation.

It is recommended to avoid asking about the personal experience of participants as they may misrepresent their true beliefs and give the answer they think that is expected of good caregivers. It may be better to ask about the experience or beliefs of people in their community. However, if participants wish to discuss their own beliefs, it may be possible to ask further probing questions, but not in a way that may be perceived as criticism. Similar to the one-on-one conversations, it is important to minimize notetaking and to consider a voice recorder (with the participant's permission) to capture the conversations.

Note: The text presented in upper case letters in the questionnaire serves as instructions for the interviewer, whereas the text in lower case letters are the questions to be asked when interviewing participants.

At the end of each subsection that investigates a potential cause for a low yellow fever vaccination coverage, a summary line allows the user of the questionnaire to review the gathered data and identify if there was a gap on this topic and, if 'yes', at which level it occurred (Box 3).

Box 3. Example of a summary line to review data.

B.1.1 Inaccuracy of estimation of yellow fever vaccination coverage

DO THE DENOMINATOR ESTIMATES FOR ADMINISTRATIVE COVERAGE COME FROM:

- | | | |
|--|------------------------------|-----------------------------|
| • CENSUS DATA (IF YES, GIVE THE YEAR WHEN THE CENSUS WAS CONDUCTED)? | YES ☐ | NO ☐ |
| • PROJECTED POPULATION ESTIMATES? | YES ☐ | NO ☐ |
| • ANOTHER SOURCE (IF YES, SPECIFY)? | YES ☐ | NO ☐ |

ARE THESE DATA ON THE IMMUNIZATION TARGET POPULATION USED SIMILAR TO THOSE FROM OTHER SOURCES:

- | | | | |
|-----------------------------------|------------------------------|-----------------------------|--------------------------|
| • REGARDING ROUTINE IMMUNIZATION? | YES ☐ | NO ☐ | IF YES, GO TO QUESTION 8 |
| • REGARDING MASS VACCINATION? | YES ☐ | NO ☐ | |

Structure of questionnaire:

A. Yellow fever vaccination coverage estimates in the district

A.1. Yellow fever vaccination coverage for routine immunization for the past two years

A.2 Yellow fever vaccination coverage after the mass vaccination campaigns conducted in the district for the past 20 years

B. Evaluation of the root causes of low yellow fever vaccination coverage

B.1. Policy-based reasons

B.1.1 Inaccuracy of estimation of yellow fever vaccination coverage

B.1.2 Gap in the vaccination plan for the general population and populations at higher risk for yellow fever

B.2 Provider-based reasons

B.2.1 Missed routine vaccination sessions

B.2.2 Stock-out of yellow fever vaccines/vaccination equipment during routine immunization

B.2.3 Lack of qualified vaccinators for routine immunization

B.2.4 Lack of knowledge/misconceptions

B.2.5 Other reasons for not vaccinating against yellow fever in routine activities

B.2.6 Other reasons for not being vaccinated against yellow fever during the mass vaccination campaign(s) implemented in the district

B.3 User-based reasons

B.3.1 Vaccine hesitancy

B.3.2 Yellow fever vaccine knowledge and misconceptions

B.3.3 Obstacles to the vaccination activities (routine and campaigns)

B.3.4 Lack of awareness of yellow fever vaccination campaigns

QUESTIONNAIRE

Name of the district evaluated _____

SECTION A. Yellow fever vaccination coverage estimates in the district

A.1. Yellow fever vaccination coverage for routine immunization for the past two years

Purpose: This section reports national and subnational yellow fever vaccination coverage results by data sources.

Data sources for finding information:



Country profile



Immunization records on administrative vaccination coverage of yellow fever vaccine through routine immunization



International Coordinating Group on Vaccine Provision (yellow fever) application form



WHO/United Nations Children's Fund (UNICEF) estimates of national vaccination coverage data ([link](#))



Published articles on vaccination



Weekly Epidemiological Record

1

Report routine yellow fever vaccination coverage over the last 2 years in the table below. In the absence of coverage data, mark as 'not available'.

DR	Level	Years	Source of results (administrative yellow fever coverage, etc.)	Date when vaccination coverage was calculated (month/year [MM/YYYY])	Yellow fever vaccination coverage (%)			
					Overall	Girls	Boys	12-23 months
	National	Current year						
		Current year -1						
		Current year - 2						
	District	Current year						
		Current year -1						
		Current year- 2						

DR	Level	Years	Source of results (administrative yellow fever coverage, etc.)	Date when vaccination coverage was calculated (month/year [MM/ YYYY])	Yellow fever vaccination coverage (%)			
					Overall	Girls	Boys	12-23 months
		Current year						
	[NAME 1]	Current year -1						
		Current year- 2						
	[NAME 2]	Current year						
		Current year -1						
		Current year- 2						
	[NAME 3]	Current year						
		Current year -1						
		Current year- 2						
	[NAME 4]	Current year						
		Current year -1						
		Current year- 2						
	[NAME 5]	Current year						
		Current year -1						
		Current year- 2						
2		USING THE INFORMATION GATHERED ABOVE, IDENTIFY THE HEALTH CENTRE OFFERING VACCINATION SESSIONS THAT BELONG TO THE HEALTH AREA WITH THE LOWEST YELLOW FEVER VACCINATION COVERAGE IN THE DISTRICT. CONDUCT INTERVIEWS IN THIS HEALTH CENTRE TO COMPLETE THE EPI HC SECTION.						
		Name of the health centre where the interview is conducted: _____						

A.2 Yellow fever vaccination coverage after the mass vaccination campaigns conducted in the district for the past 20 years

Purpose: This section reports vaccination coverage results during yellow fever mass vaccination campaigns if they have taken place in the district.

Data sources for finding information:



Country profile



Immunity gap tool



Reactive vaccination campaign reports



Preventive vaccination campaign reports



International Coordinating Group on Vaccine Provision (yellow fever) application form



WHO/United Nations Children's Fund (UNICEF) estimates of national vaccination coverage data [link](#)



Demographic health surveys



Vaccination articles published



Weekly Epidemiological Record

DR	3	Were there any mass campaigns of yellow fever in the district?		Yes	No	If NO, go to question 5						
	4	If YES, list and describe below the different yellow fever mass campaigns (reactive [RVC] and/or preventive [PMVC]) that occurred in the district for the past 20 years. ¹										
RVC/ PMVC	Date range of the mass campaigns (MM/YYYY-MM/YYYY)	Age group targeted by the campaigns	Method used to calculate vaccination coverage (administrative/cluster/other)	Vaccination coverage with recall by individual/caregiver or with vaccination records	Yellow fever vaccination coverage (%)							
					Overall	Women	Men	<1 year	1- <5 years	5- <15 years	≥ 15 years	

¹ There are different methodologies for measuring vaccination coverage that can be used either to monitor the vaccination campaign (conducted during or after the campaign) or to assess vaccination coverage after the implementation of the campaign. Guidance on the use of these methodologies is presented in chapter 5 of the WHO Eliminate yellow fever epidemics (EYE) strategy country toolkit (2024) (4).

SECTION B. Evaluation of the root causes of low yellow fever vaccination coverage

B.1. Policy-based reasons

Purpose: These questions assess the specific health policies that may have led to the failure to vaccinate with the yellow fever vaccine.

B.1.1 Inaccuracy of estimation of yellow fever vaccination coverage

Data sources for finding information:



Vaccination procedure regulations (current immunization protocol standard, current national vaccination strategic plan)



Vaccination records at the health centre level (health centre offering vaccination sessions and belonging to the health area with the lowest yellow fever vaccination coverage)



Interviews with the vaccinators working in the health centre offering vaccination sessions and belonging to the health area with the lowest yellow fever vaccination coverage

DR	5	Do the denominator estimates for administrative coverage come from:			
		• census data (if YES, give the year when the census was conducted)?	Yes ☐	No ☐	
		• projected population estimates?	Yes ☐	No ☐	
		• another source (if YES, specify)?	Yes ☐	No ☐	
	6	Are these data on the immunization target population used similar to those from other sources. ²			
		• regarding routine immunization?	Yes ☐	No ☐	If YES, go to question 8
		• regarding mass vaccination?	Yes ☐	No ☐	
	7	If NO, why not?			
	8	Is the denominator used for the vaccination coverage accurate:			
		• regarding routine immunization?	Yes ☐	No ☐	
		• regarding mass vaccination?	Yes ☐	No ☐	

² The denominators of administrative vaccination coverage are based on population estimates that may be outdated (for example, due to old censuses). It is therefore important to cross-reference data from the target population for routine immunization with other sources of data on births (birth registries and family planning registries) as outlined in the WHO manual 'Assessing and improving the accuracy of the estimated target population for immunization coverage (5)', and in the WHO Measles outbreak guide 2022 (3), or on the general population for mass vaccination campaigns (for example., target population for the malaria programme data).

	9	Is the numerator used for the vaccination coverage accurate:			
		• regarding routine immunization?	Yes ☐	No ☐	If YES, go to question 11
		• regarding mass vaccination?	Yes ☐	No ☐	
	10	If NO, what are the reasons:			
		• data entry errors in the tally sheet?	☐		
		• inaccurate recording of the count of vaccinated persons from the tally sheet to the synthesis report at the health centre level?	☐		
		• other (DESCRIBE)?	☐		
	11	LOOK AT QUESTION 8 AND 9. IF YOU ANSWER NO TO AT LEAST ONE OF THESE QUESTIONS, CONTINUE TO QUESTION 12, OTHERWISE PROCEED TO QUESTION 16.			
	12	Did any of these gaps lead to an overestimation of the calculation of administrative vaccination coverage of routine or mass activities?	Yes ☐	No ☐	If NO, go to question 14
	13	If YES, describe.			
	14	Did any of those gaps, lead to an underestimation of the calculation of administrative vaccination coverage of routine or mass activities?	Yes ☐	No ☐	If NO, go to question 16
	15	If YES, describe.			
EPI HC	16	REVIEW THE IMMUNIZATION RECORDS AT HEALTH CENTRE LEVEL WITH THE VACCINATORS. Are there any discrepancies in the number of children vaccinated between the different data collection sheets and/or between the collection sheets and the document reporting the results to the central level (District Health Information Software [DHIS2], etc.)?	Yes ☐	No ☐	If NO, go to question 18
	17	If YES, describe the reasons.			
SUMMARY	18	REVIEW THE RESPONSES TO QUESTIONS 5 TO 17. IS THERE A GAP IN THE ESTIMATION IN THE CALCULATION OF THE YELLOW FEVER VACCINATION COVERAGE, WHICH MAY HAVE UNDERESTIMATED THE RISK OF AN OUTBREAK AND PREVENTED THE IMPLEMENTATION OF APPROPRIATE VACCINATION STRATEGIES:			
		IN THE ROUTINE VACCINATION?	YES ☐	NO ☐	IF NO, GO TO QUESTION 20
		IN THE MASS VACCINATION CAMPAIGN?	YES ☐	NO ☐	
	19	IF YES, AT WHICH LEVEL DID THE GAP OCCUR?			
		• CENTRAL LEVEL?	☐		
		• HEALTH CENTRE LEVEL?	☐		

B.1.2 Gap in the vaccination plan for the general population and populations at higher risk for yellow fever.³

Data sources for finding information:



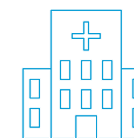
Vaccination procedure regulations (current immunization protocol standard, current national vaccination strategic plan)



Immunization records at the health centre level (health centre offering vaccination sessions and belonging to the health area with the lowest yellow fever vaccination coverage)



Interviews with the district EPI focal point/supervisor



Interviews with the vaccinators working in the health centre offering vaccination sessions and belonging to the health area with the lowest yellow fever vaccination coverage

DR	20	What is the age range during which children can receive the yellow fever vaccine as part of the routine activity?			
	21	Are catch-up vaccination sessions available if the child did not get the yellow fever vaccine in this age group?	Yes ☐	No ☐	If NO, go to question 23
	22	If YES, up to what age?			
	23	Is there a yellow fever catch-up vaccination strategy for a newly-arrived population at higher risk in one district (either with occupational risk factors or due to sociocultural reasons, for example, marginalized populations, zero dose communities, mobile population)?	Yes ☐	No ☐	If NO, go to question 25
	24	If YES, describe the vaccination strategy (SINCE WHEN, POPULATIONS AT HIGHER RISK, ETC.).			
	25	Do the children aged (AGE RANGE INDICATED IN THE ANSWER TO QUESTION 20) in the population at higher risk or seasonal workers have access to the country's EPI services?	Yes ☐	No ☐	If NO, go to question 27
	26	If YES, describe.			

³ For the definition of population at higher risk for yellow fever refer to the WHO yellow fever vaccine-preventable diseases surveillance standards manual (2020) (6).

EPI D	27	Are there population groups at higher risk of yellow fever in the district?	Yes ☐	No ☐	If NO, go to question 30
	28	If YES, are you implementing yellow fever catch-up vaccination strategies for newly arrived populations at higher risk? (TO BE CHECKED WITH CENTRAL LEVEL QUESTION 23 RESPONSE.)	Yes ☐	No ☐	If NO, go to question 30
	29	If YES, describe the vaccination strategy.			
	30	Do the children aged (AGE RANGE INDICATED IN THE ANSWER TO QUESTION 20) in the population at higher risk or seasonal workers have access to the EPI? (TO BE CHECKED WITH CENTRAL LEVEL QUESTION 25 RESPONSE.)	Yes ☐	No ☐	
EPI HC	31	Are there population groups at higher risk in the health area?	Yes ☐	No ☐	If NO, go to question 35
	32	If YES, are you implementing yellow fever catch-up vaccination strategies for newly arrived populations at higher risk? (TO BE CHECKED WITH CENTRAL LEVEL QUESTION 23 RESPONSE.)	Yes ☐	No ☐	If NO, go to question 34
	33	If YES, describe the vaccination strategy.			
	34	Do the children aged (AGE RANGE INDICATED IN THE ANSWER TO QUESTION 20) in the population at higher risk or seasonal workers have access to the EPI services? (TO BE CHECKED WITH THE CENTRAL LEVEL QUESTION 25 RESPONSE.)	Yes ☐	No ☐	
SUMMARY	35	REVIEW THE RESPONSES TO QUESTIONS 20 TO 34. BASED ON THE INFORMATION PROVIDED, ARE THERE ANY GAPS IN THE NATIONAL VACCINATION PLAN THAT MAY HAVE CONTRIBUTED TO MISSED OPPORTUNITIES FOR YELLOW FEVER VACCINATION DURING ROUTINE ACTIVITIES?	YES ☐	NO ☐	IF NO, GO TO QUESTION 37
	36	IF YES, AT WHICH LEVEL DID THE GAP OCCUR:			
		CENTRAL LEVEL? DISTRICT LEVEL? HEALTH CENTRE LEVEL?	☐ ☐ ☐		

B.2. Provider-based reasons

Purpose: These questions assess the missed opportunities of vaccination that may have led to a failure to vaccinate with the yellow fever vaccine.

B.2.1 Missed routine vaccination sessions

Data sources for finding information:



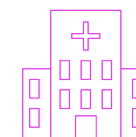
Vaccination reports of the district



Catch-up vaccination guidance (if available)



Interviews with the district EPI focal point/supervisor



Interviews with the director and vaccinators working at the health centre offering vaccination sessions activities belonging to the health area with the lowest yellow fever vaccination coverage.

DR & EPI DO	37	FAMILIARIZE YOURSELF WITH THE DISTRICT AND ITS CATCHMENT AREA ON A MAP.			
		ASK FOR THE NECESSARY REPORTS TO COMPLETE THE TABLE BELOW REGARDING THE VACCINATION SESSIONS (STATIC OR OUTREACH/MOBILE) <u>PLANNED AND COMPLETED</u> .			
		IF A SESSION DID NOT TAKE PLACE, CALCULATE THE NUMBER OF MISSED SESSIONS AND ASK THE DISTRICT EPI FOCAL POINT WHY THIS HAPPENED.			
DR & EPI DO	38	Complete the table			
			Health areas with low yellow fever vaccination coverage (<80%)	Health areas with high yellow fever vaccination coverage (≥ %80)	Total
	Number of children eligible for routine immunization (AGE RANGE INDICATED IN QUESTION 20)	Current year			
		Current year -1			
	Static sessions				
	Number of static sessions <u>planned per month</u> (=a)	Current year			
		Current year -1			

Total number of static sessions planned per year (b)= $a \times 12$	Current year
	Current year -1
Total number of static sessions <u>conducted in the year under review</u> (c)	Current year
	Current year -1
Percentage of planned static sessions conducted in the year under review ($c/b \times 100$)	Current year
	Current year -1
Outreach and mobile sessions	
Number of outreach/mobile sessions planned per month (d)	Current year
	Current year -1
Total number of outreach/mobile sessions <u>planned per year</u> (e)= $d \times 12$	Current year
	Current year -1
Total number of outreach/mobile sessions <u>conducted in the year under review</u> (=f)	Current year
	Current year -1
Percentage of planned outreach/mobile sessions conducted in the year under review ($f/e \times 100$)	Current year
	Current year -1
Total number of sessions	
Total number of <u>planned sessions per year</u> (b+e)	Current year
	Current year -1
Total number of sessions <u>conducted per year</u> (c+f)	Current year
	Current year -1
Percentage of all sessions <u>conducted in the year under review</u> ($(c/b \times 100) + (f/e \times 100)$)	Current year
	Current year -1

	39	What are the reasons why the sessions did not take place: (RECORD ALL THAT APPLY.)			
		• lack of staff?	☐		
		• lack of yellow fever vaccines?	☐		
		• lack of transportation?	☐		
		• hard-to-reach area?	☐		
		• security?	☐		
		• other (DESCRIBE)?	☐		
EPI HC	40	Have there been any missed vaccination sessions:			
		• Last year?	Yes ☐	No ☐	
		• This year?	Yes ☐	No ☐	If NO, go to question 45
	41	If YES, were they:			
		• static sessions?	☐		
		• outreach sessions?	☐		
		• mobile sessions?	☐		
	42	If YES, how many were missed:			
		• last year?			
		• this year?			
	43	What were the reasons for this: (CHECK ALL THAT APPLY)			
		• lack of staff?	☐		
		• lack of yellow fever vaccines?	☐		
		• lack of transportation?	☐		
		• hard-to-reach area?	☐		
		• security?	☐		
		• other (DESCRIBE)?	☐		
	44	How can the problem be solved?			

SUMMARY	45	REVIEW THE RESPONSES TO QUESTIONS 37 TO 44. WERE THERE ANY MISSED VACCINATION SESSIONS IN THE DISTRICT IN THE LAST 2 YEARS? YES ☐ NO ☐																																																									
DR & EPI D	46	REVIEW THE DISTRICT'S EPI VACCINATION SUMMARY REPORT AND COMPLETE THE TABLE BELOW ON DROP-OUT RATES.																																																									
Drop-out rate in the district <table border="1"> <thead> <tr> <th>Year</th> <th>A</th> <th>B</th> <th>C</th> <th>D</th> <th>E</th> <th>F</th> <th>G</th> <th>H</th> <th>I</th> </tr> </thead> <tbody> <tr> <td></td> <td colspan="9" style="text-align: right;">Drop-out rate between:</td> </tr> <tr> <td></td> <td>Number of registered children</td> <td>Number of BCG*</td> <td>Number of Penta 1*</td> <td>Number of MCV1*</td> <td>Number of YFV*</td> <td>Registered children and YFV</td> <td>BCG and YFV</td> <td>Penta 1 and YFV</td> <td>MCV1 and YFV</td> </tr> <tr> <td>Current year -1**</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Current year</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table> * BCG: bacillus Calmette-Guérin; Penta 1: pentavalent first injection; MCV1: measles-containing-vaccine first-dose; YFV: yellow fever vaccine. ** Assuming MCV1 schedule is 9-11 months.										Year	A	B	C	D	E	F	G	H	I		Drop-out rate between:										Number of registered children	Number of BCG*	Number of Penta 1*	Number of MCV1*	Number of YFV*	Registered children and YFV	BCG and YFV	Penta 1 and YFV	MCV1 and YFV	Current year -1**										Current year									
Year	A	B	C	D	E	F	G	H	I																																																		
	Drop-out rate between:																																																										
	Number of registered children	Number of BCG*	Number of Penta 1*	Number of MCV1*	Number of YFV*	Registered children and YFV	BCG and YFV	Penta 1 and YFV	MCV1 and YFV																																																		
Current year -1**																																																											
Current year																																																											
	47	REVIEW THE RESULTS WITH THE DISTRICT EPI FOCAL POINT AND ASK THE FOLLOWING QUESTIONS IF APPLICABLE. IF NOT APPLICABLE GO TO QUESTION 51. Why is the number of children who received the yellow fever vaccine lower than the number of children registered and/or lower than the number of children who received BCG?																																																									
	48	How can this problem be addressed?																																																									
	49	Why is the number of children who received the yellow fever vaccine lower than the number of children who received the MCV1?																																																									
	50	How can this problem be addressed?																																																									
SUMMARY	51	REVIEW THE RESPONSES TO QUESTIONS 46 TO 50. WERE THERE ANY GAPS IN COMMUNITY ADHERENCE WITH THE ROUTINE IMMUNIZATION ACTIVITY IN THE LAST 2 YEARS? YES ☐ NO ☐																																																									

B.2.2 Stock-out of yellow fever vaccines/vaccination equipment during routine immunization

Data sources for finding information:



Vaccine books and purchase orders at district level, including the health centre offering vaccination sessions and belonging to the health area with the lowest yellow fever vaccination coverage



Interviews with the logistician of the vaccine storage room at district level and at the health centre offering vaccination sessions and belonging to the health area with the lowest yellow fever vaccination coverage

DR and Log D	52	REVIEW WITH THE MANAGER OF THE VACCINE STORAGE ROOM THE YELLOW FEVER VACCINES' STOCK BOOK AND THE PURCHASE ORDERS						
		Have there ever been stock-outs of yellow fever vaccines and/or vaccination equipment at the district level for routine immunization?				Yes ☐	No ☐	If NO, go to question 57
	53	If YES, complete the table.						
		Item out of stock (for example, yellow fever vaccines, syringes, etc.)	Year (YYYY) of stock shortage	Duration of stock shortage	Reason for stock shortage (for example, poor planning of needs, expired vaccines, etc.)			
	54	If YES, has this caused vaccination teams to cancel or shorten planned vaccination sessions?				Yes ☐	No ☐	If NO, go to question 57
	55	If YES, provide details.						
	56	How can the problem be solved?						

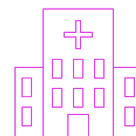
DR and Log HC	57	Have there ever been stock-outs of yellow fever vaccines and/or vaccination equipment in your health centre for routine immunization?				Yes ☐	No ☐	If NO, go to question 63
	58	If YES, complete the table.						
	59	Item out of stock (for example, yellow fever vaccines, syringes, etc.)	Year (YYYY) of stock shortage	Duration of stock shortage	Reason for stock shortage (for example, poor planning of needs, expired vaccines, etc.)			
	60	If YES, has this caused the vaccination team to cancel or shorten planned vaccination sessions?				Yes ☐	No ☐	If NO, go to question 63
	61	If YES, provide details.						
	62	How can the problem be solved?						
SUMMARY	63	REVIEW THE RESPONSES TO QUESTIONS 52 TO 62.						
		WAS THERE A YELLOW FEVER VACCINE STOCKOUT THAT LED TO THE CANCELLATION OR THE SHORTENING OF ROUTINE VACCINATION SESSIONS?				YES ☐	NO ☐	IF NO, GO TO QUESTION 65
	64	IF YES, AT WHICH LEVEL DID THE STOCK-OUT OCCUR:						
		• DISTRICT LEVEL?				☐		
		• HEALTH CENTRE LEVEL?				☐		

B.2.3 Lack of qualified vaccinators for routine immunization

Data sources for finding information:



Interviews with the district EPI focal point/supervisor



Interviews with the director and vaccinators working at the health centre offering vaccination sessions and belonging to the health area with the lowest yellow fever vaccination coverage

EPI D	65	How many health centres offer vaccination sessions in the district this year?			
	66	How many sanctioned vaccinators are there in the district this year?			
	67	How many sanctioned vaccinators are there on average per health centre offering vaccination sessions in the district this year (= QUESTIONS 66/65)?			
	68	How many children aged (AGE RANGE INDICATED IN QUESTION 20) per one sanctioned vaccinator are there on average in the district this year (= QUESTIONS 38/66)?			
	69	Does the number of sanctioned vaccinators meet the recommended standard?			
	70	From a supervisor's perspective, do you think that the number of <u>sanctioned vaccinators</u> working in the district is sufficient to conduct all planned vaccination sessions?	Yes ☐	No ☐	If YES, go to question 72
	71	If NO, what are the reasons why the number of <u>sanctioned vaccinators</u> in the district is not sufficient: (LIST THE FOLLOWING ANSWERS. TICK ALL THE ANSWERS THAT SUPERVISORS CONSIDER RELEVANT.)			
		- no funds for recruitment? - not enough training? - other (DESCRIBE)?	☐	☐	☐
72	Has the lack of <u>sanctioned vaccinators</u> prevented the vaccination team from organizing routine vaccination activities in the district?	Yes ☐	No ☐	If NO, go to question 74	
73	If YES, in which year?				

EPI HC	74	How many positions of sanctioned vaccinators are there working in the health centre to cover your health area?			
	75	Are there other positions for vaccinators (volunteers) in your health centre?	Yes ☐	No ☐	
	76	How many vaccinators do you have in total?			
	77	Do you think that this number of sanctioned vaccinators is sufficient to conduct all the vaccination sessions planned in your health area?	Yes ☐	No ☐	If YES, go to 80
	78	If NO, explain why.			
	79	If NO, why are there not more vaccinators: (LIST THE FOLLOWING ANSWERS. TICK ALL THE ANSWERS THAT THE IMMUNIZATION PROVIDERS CONSIDER RELEVANT.)			
		• no funds for recruitment? • not enough training? • other (DESCRIBE)?			
	80	Has a lack of vaccinators ever prevented the vaccination team from conducting routine vaccination sessions?	Yes ☐	No ☐	If NO, go to 82
	81	If YES, which year?			
SUMMARY	82	REVIEW THE RESPONSES TO QUESTIONS 65 TO 81.			
	83	WAS THERE A LACK OF VACCINATORS THAT LED TO THE CANCELLATION OF ROUTINE VACCINATION SESSIONS?	YES ☐	NO ☐	

B.2.4 Lack of knowledge/misconceptions

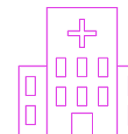
Data sources for finding information:



Vaccination procedure regulations (current immunization protocol standard, current national vaccination strategic plan)



Interviews with the district EPI focal point/supervisor



Interviews with the director and vaccinators working at the health centre offering vaccination session and belonging to the health area with the lowest yellow fever vaccination coverage

a. Regarding yellow fever vaccination policy

Failure to open vials if a certain number of children are not present at the routine vaccination session

DR	84	For how many children can a vial of yellow fever vaccines be opened during a vaccination session?			
EPI D	85	How many children have to be present before the vaccinator opens a yellow fever vaccine vial? (CHECK IF THE ANSWER IS IN LINE WITH THE OPENING POLICY - QUESTION 84.)			
EPI HC	86	Do you open a vial of yellow fever vaccine if there is only one child present at the vaccination session? (CHECK IF THE ANSWER IS IN LINE WITH THE OPENING POLICY QUESTION 84.).	Yes ☐	No ☐	If YES, go to question 89
	87	If NO, how many children must be present before a yellow fever vaccine vial is opened?			
	88	If NO, what do you say to the caregiver of the child who came to be vaccinated for yellow fever and is the only child present at the vaccination session?			

Vaccination schedule of the routine immunization activity

EPI D	89	If a caregiver brings a child over (AGE RANGE INDICATED IN QUESTION 20) months to be vaccinated against yellow fever, should the vaccinator administer the vaccine?	Yes ☐	No ☐	If NO, go to question 91
	90	If YES, up to what age can the vaccinator administer the yellow fever vaccine? (TO BE CHECKED WITH QUESTION 22 WHEN THE COUNTRY CONDUCT CATCH-UP VACCINATION SESSION.)			Go to question 92

	91	If NO, what are the reasons: (LIST THE FOLLOWING ANSWERS. TICK ALL THE ANSWERS THAT THE SUPERVISOR CONSIDERS RELEVANT.) - the vaccination schedule does not authorize to administer the yellow fever vaccine after 11 months? ☐ - this creates problems for recording data in the vaccination file? ☐ - administration of the yellow fever vaccine after 11 months leads to vaccine failure? ☐ - I consider vaccinating children >11 months of age to be a waste of resources? ☐ - other (DESCRIBE)? ☐			
EPI HC	92	If a caregiver brings a child over (AGE RANGE INDICATED IN QUESTION 20) months of age to be vaccinated against yellow fever, do you administer the vaccine?	Yes ☐	No ☐	If NO, go to question 94
	93	If YES, up to what age do you administer the yellow fever vaccine? TO BE CHECKED WITH QUESTION 22 WHEN THE COUNTRY CONDUCT CATCH-UP VACCINATION SESSION.			Go to question 95
	94	If NO, what are the reasons: (LIST THE FOLLOWING ANSWERS. TICK ALL THE ANSWERS THAT THE VACCINATOR CONSIDERS RELEVANT.) - the vaccination schedule does not allow to administer the yellow fever vaccine after 11 months? ☐ - this creates problems for recording data in the vaccination file? ☐ - administration of the yellow fever vaccine after 11 months results in vaccine failure? ☐ - I consider vaccinating children >11 months of age to be a waste of resources? ☐ - other (DESCRIBE)? ☐			
Access to the routine immunization service					
DR	95	According to the vaccination policy of the country, can children eligible for the routine EPI be vaccinated against yellow fever: - in all health facilities in the country? ☐ - in specific healthcare facilities according to their area of residence? ☐			
EPI HC	96	What do you do if a mother comes with her child to be vaccinated against yellow fever during a routine vaccination session, even though they do not reside in the health area? (TO BE CHECKED WITH QUESTION 95.)			

b. Regarding the co-administration of the yellow fever vaccine with another antigen (during routine activities and mass vaccination campaigns)

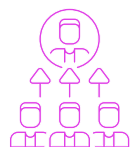
DR	97	Is there a policy of co-administration of the yellow fever vaccine with other antigens (for example, measles) during routine activities, or mass vaccination campaigns?	Yes ☐	No ☐
EPI D	98	Is it possible to co-administer the yellow fever vaccine at the same time as another vaccine? (TO BE CHECKED WITH QUESTION 97.)	Yes ☐	No ☐
EPI HC	99	Is it possible to administer the yellow fever vaccine at the same time as another vaccine? (TO BE CHECKED WITH QUESTION 97.)	Yes ☐	No ☐

c. Regarding the contraindications of the yellow fever vaccine (during routine activities and mass vaccination campaigns)

EPI HC	100	Can a child receive a yellow fever vaccine if the child has:	
		• a mild respiratory illness (for example, cough, runny nose)?	☐
		• diarrhea?	☐
		• proven recent contact with a sick person?	☐
		• previous local reaction from vaccination (see list below):	☐
		– redness	☐
		– swelling	☐
		– low fever	☐
		– crying	☐
		– soreness, pain	☐
		– only if two or more of the above?	☐
		• egg allergy?	☐
		• previous anaphylactic or severe allergic reaction to vaccine?	☐
		• severe illness (for example., high fever)?	☐
		• child is breastfeeding?	☐

SUMMARY	101	REVIEW THE RESPONSES TO QUESTIONS 84 TO 100.			
		WAS THERE A LACK OF KNOWLEDGE OR MISCONCEPTION ABOUT THE VACCINATION POLICY AND/OR THE YELLOW FEVER VACCINE AMONG VACCINATORS THAT RESULTED IN A MISSED OPPORTUNITY FOR YELLOW FEVER VACCINATION?	YES ☐	NO ☐	IF NO, GO TO QUESTION 132
	102	IF YES, WAS THE MISCONCEPTION RELATED TO:			
		• The vaccination policy?	☐		
		• The yellow fever vaccine?	☐		
	103	IF YES, AT WHICH LEVEL DID THE GAP OCCUR:			
		• Central level?	☐		
		• District level?	☐		
		• Health centre level?	☐		

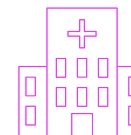
Data sources for finding information:



Record of supervisory visits



Interviews with the district EPI focal point/
supervisor



Interviews with the director and vaccinators
working at the health centre offering vaccination
sessions and belonging to the health area with the
lowest yellow fever vaccination coverage

Limitation in skills enhancement

EPI D	104	Are there any vaccine policy guidelines available in health centres for vaccinators?	Yes ☐	No ☐	If NO, go to question 106
	105	If YES, since when?			
	106	What types of training do vaccinators receive?			

	107	How often are they trained:			
		• annually?	☐		
		• quarterly?	☐		
		• ad hoc?	☐		
		• other (DESCRIBE)?	☐		
	108	Do you think that vaccinators working in district health centres need <u>additional training on immunization</u> to do their work properly?	Yes ☐	No ☐	If NO, go to question 110
	109	If YES, what training(s) would be required?			
	110	Did the absence of this training(s) lead to any specific problems?	Yes ☐	No ☐	If NO, go to question 113
	111	If YES, describe the problems this has caused.			
	112	If YES, how can the problem be solved?			
EPI HC	113	As a vaccinator, would you need <u>additional training in immunization</u> to perform your duties?	Yes ☐	No ☐	If NO, go to question 115
	114	If YES, what training would be required?			
	115	Did the absence of this training lead to any specific problems?	Yes ☐	No ☐	If NO, go to question 118
	116	If YES, describe the problems this has caused.			
	117	If YES, how can the problem be solved?			
Lack of supervision in vaccination activities					
EPI D	118	How many <u>sanctioned supervisor positions</u> were there in the district:			
		• last year?			
		• this year?			
	119	How many <u>have been filled</u> :			
		• last year?			
		• this year?			
	120	How many health centres offering vaccination sessions are there per supervisor (= QUESTIONS 65/119)?			
	121	Do you think that the number of sanctioned supervisors working in the district is enough to complete all planned supervision sessions?	Yes ☐	No ☐	If YES, go to question 123

	122	If NO, what are the reasons why the number of <u>sanctioned supervisors</u> in the district is not sufficient: (LIST THE FOLLOWING ANSWERS. TICK ALL THE ANSWERS THAT THE SUPERVISOR CONSIDERS RELEVANT.)			
		• no funds for recruitment?	☐		
		• not enough training?	☐		
		• other (DESCRIBE)?	☐		
	123	Have there been any problems that had made it difficult to supervise routine immunization activities in health facilities offering vaccination sessions?	Yes ☐	No ☐	If NO, go to question 126
	124	If YES, what were they:			
		• transport unavailable?	☐		
		• limited budget?	☐		
		• hard-to-reach areas (security, geographical barriers, etc.)?	☐		
		• competing priorities (for example, Covid-19-related activities)?	☐		
		• other (DESCRIBE)?	☐		
	125	If YES, how can this problem be solved?			
EPI HC	126	How many times has the district supervisor visited your health centre:			
		• last year?	☐		
		• this year?	☐		
	127	Do you think that the supervision of vaccinators and vaccination sessions is adequate?	Yes ☐	No ☐	If YES, go to question 130
	128	If NO, why is it inadequate?			
	129	If NO, what can be done to improve the supervision?			

SUMMARY	130	REVIEW THE RESPONSES TO QUESTIONS 104 TO 129.	
		WAS THE KNOWLEDGE GAP IDENTIFIED IN QUESTION 101 ATTRIBUTABLE TO:	
		LIMITATIONS IN SKILLS DEVELOPMENT?	☐
		INSUFFICIENT SUPERVISION OF VACCINATION ACTIVITIES?	☐
	131	AT WHICH LEVEL DID THE GAP OCCUR?	
		CENTRAL LEVEL?	☐
		DISTRICT LEVEL?	☐
		HEALTH CENTRE LEVEL?	☐

B.2.5 Other reasons for not vaccinating against yellow fever in routine activities

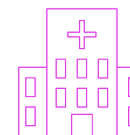
Data sources for finding information:



Vaccination reports of the district



Interviews with the district EPI focal point/
supervisor



Interviews with the director and vaccinators
working at the health centre offering vaccination
sessions and belonging to the health area with the
lowest yellow fever vaccination coverage

EPI D	132	Are there any other issues that you can think of that have cancelled routine vaccination activities during the two past years?	Yes ☐	No ☐	If NO, go to question 134
	133	If YES, which ones?			
EPI HC	134	Are there any other issues you can think of that have cancelled routine vaccination activities during the two past years?	Yes ☐	No ☐	If NO, go to question 137
	135	If YES, which ones?			
	136	If YES, how can this problem be solved?			

SUMMARY	137	REVIEW THE RESPONSES TO QUESTIONS 133 TO 137.			
		WERE THERE ADDITIONAL FACTORS THAT HAVE CONTRIBUTED TO VACCINATION MISSED OPPORTUNITIES DURING ROUTINE SESSIONS?	YES ☐	NO ☐	IF NO, GO TO QUESTION 139
	138	IF YES, DESCRIBE ALL AND INDICATE AT WHICH LEVEL THE GAP OCCURRED:			
		• CENTRAL LEVEL?	☐		
		• DISTRICT LEVEL?	☐		
		• HEALTH CENTRE LEVEL?	☐		

B.2.6 Other reasons for not being vaccinated against yellow fever during the mass vaccination campaign(s) implemented in the district

Data sources for finding information:



Interviews with the district EPI focal point/
supervisor

DR	139	Was there one or several yellow fever vaccination campaigns implemented in the district in the last 20 years (REFER TO QUESTION 3) ?	Yes ☐	No ☐	If NO, go to question 147
	140	If YES, did any vaccination campaigns record a vaccination rate below 80%? (4)	Yes ☐	No ☐	If NO, go to question 147
EPI D	141	PROCEED WITH THE QUESTIONS BELOW FOR EACH VACCINATION CAMPAIGN THAT REPORTED LESS THAN 80% YELLOW FEVER VACCINE COVERAGE. Were there any areas in the district that were not reached by the vaccination campaign?	Yes ☐	No ☐	If NO, go to question 143

142	If YES, what were the causes: (TICK ALL THE ANSWERS THAT THE SUPERVISOR CONSIDERS RELEVANT.) - insufficient number of vaccination teams? ☐ - difficulty to access to certain areas? ☐ - security issues? ☐ - lack of microplanning? ☐ - vaccine stock-out during the campaign? ☐ - cold chain issues? ☐ - other (DESCRIBE)? ☐			
143	Was a communication and social mobilization plan for the mass campaign implemented in the district?	Yes ☐	No ☐	If YES, go to question 145
144	If NO, explain the reasons.			
SUMMARY	145 REVIEW THE RESPONSES TO QUESTIONS 139 TO 144. WERE THERE ADDITIONAL FACTORS THAT CONTRIBUTED TO VACCINATION MISSED OPPORTUNITIES DURING THE MASS VACCINATION CAMPAIGNS? YES ☐ NO ☐ IF NO, SKIP TO QUESTION 147			
146	IF YES, DESCRIBE WHICH ONE(S) AND AT WHICH LEVEL THE GAP OCCURRED? CENTRAL LEVEL? ☐ DISTRICT LEVEL? ☐ HEALTH CENTRE LEVEL? ☐			

B.3. User-based reasons

Purpose: These questions assess the user-based reasons that may have led to a failure to vaccinate with the yellow fever vaccine.

B.3.1 Vaccine hesitancy

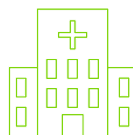
Data sources for finding information:



Published articles on vaccination acceptance



Interviews with the district EPI focal point/supervisor



Interviews with the director and vaccinators working at the health centre offering vaccination sessions and belonging to the health area with the lowest yellow fever vaccination coverage



Interviews with community health workers



Group discussion

DR	147	Has there been any data collected/evidence gathered on vaccine acceptance, particularly for yellow fever vaccine?	Yes ☐	No ☐	If NO, go to question 149
	148	If YES, describe the results.			
EPI D	149	Are vaccines generally well accepted in the district?	Yes ☐	No ☐	
	150	Is acceptance of yellow fever vaccine similar to other vaccines?	Yes ☐	No ☐	If YES, go to question 153
	151	If NO, why not?			
	152	If NO, how can this problem be solved?			
	153	Are you aware of any rumors, misconceptions, myths and/or beliefs (current or past) about vaccines and vaccination in general in the district?	Yes ☐	No ☐	If NO, go to question 157
	154	IF YES, please indicate the rumors, myths, beliefs and/or misconceptions.			
	155	If YES, do you remember when it was? (RECORD THE TIME FRAME IF THE PERSON DOES NOT REMEMBER THE EXACT YEAR, SUCH AS BEFORE, DURING OR AFTER THE COVID-19 PANDEMIC, ETC.)			

	156	If YES, how can these rumors, myths, beliefs and/or misconceptions be addressed?			
	157	How did the Covid-19 pandemic impact on vaccinations activities?			
	158	In your opinion, what obstacles could prevent EPI users from coming to health centres to be vaccinated?			
	159	How can this problem be solved?			
EPI HC	160	Are vaccines generally well accepted in the health area?	Yes ☐	No ☐	
	161	Is acceptance of yellow fever vaccine similar to other vaccines?	Yes ☐	No ☐	If YES, go to question 164
	162	If NO, why not?			
	163	If NO, how can this problem be solved?			
	164	Are you aware of any rumors, myths, beliefs and/or misconceptions (current or past) about vaccines and vaccination in general in the health area?	Yes ☐	No ☐	If NO, go to 168
	165	If YES, please describe the rumors, myths, beliefs and/or misconceptions.			
	166	If YES, do you remember when it was? (RECORD THE TIME FRAME IF THE PERSON DOES NOT REMEMBER THE EXACT YEAR, SUCH AS BEFORE, DURING OR AFTER THE COVID-19 PANDEMIC, ETC.)			
	167	If YES, how can these rumors, myths, beliefs and/or misconceptions be addressed?			
	168	How did the Covid-19 pandemic impact on vaccinations activities?			
	169	In your opinion, what obstacles could prevent EPI users from coming to health centres to be vaccinated during a routine session?			
	170	How can this problem be solved?			
CHW	171	How accepted are yellow fever vaccines in the health area:			
		• low?	☐		
		• moderate?	☐		
		• high?	☐		
	172	Is acceptance of the yellow fever vaccine similar to other vaccines?	Yes ☐	No ☐	If YES, go to question 175
	173	If NO, why not?			
	174	If NO, how can this problem be solved?			

	175	Are you aware of any rumors, myths, beliefs and/or misconceptions (current or past) about yellow fever vaccines in the health area?	Yes ☐	No ☐	If NO, go to question 178
	176	If YES, please indicate the rumors, myths, beliefs and/or misconceptions.			
	177	If YES, do you remember when it was? (RECORD THE TIME FRAME IF THE PERSON DOES NOT REMEMBER THE EXACT YEAR, SUCH AS BEFORE, DURING OR AFTER THE COVID-19 PANDEMIC, ETC.)			
	178	In your opinion, what obstacles could prevent EPI users from coming to health centres to be vaccinated during the routine sessions?			
GD	179	BEFORE STARTING THE DISCUSSION, LOOK AT THE PROCEDURES LISTED BEFORE THE QUESTIONNAIRE AND RECORD THE NUMBER OF PARTICIPANTS IN THE ROOM. Number of participants in the room?			
	180	Have you heard any rumors or misconceptions about yellow fever vaccine?	Number of participants that answered "YES"		
	181	If you answered YES to the questions, can you describe the rumors or misconceptions?			
	182	What was the source of the rumors or misconceptions: • social media (which one(s)? • friends/relatives/neighbors? • other (describe)?			
	183	If YES, do you believe any of these rumors or misconceptions? TRY TO ASSESS HOW MANY IN THE COMMUNITY BELIEVE ANY OF THESE RUMOURS, WHICH ONES, AND IF THEY HAVE PREVENTED THEM FROM TAKING THEIR CHILDREN TO BE VACCINATED.			
SUMMARY	184	REVIEW THE RESPONSES TO QUESTIONS 147 TO 183. IS THERE A VACCINE HESITANCY IN THE COMMUNITY THAT HAS LED TO CHILDREN NOT BEING VACCINATED?			YES ☐ NO ☐

B.3.2 Yellow fever vaccine knowledge and misconceptions

GD	185	BEFORE STARTING THE DISCUSSION, RECORD THE NUMBER OF PARTICIPANTS IN THE ROOM. Number of participants in the room?	
	186	I would like to ask you all a few questions about yellow fever disease and vaccinating children with yellow fever vaccine. Have you ever heard about yellow fever?	Number of participants who answered "YES"
	187	At what age should a child be vaccinated against yellow fever? LOOK AT THE ANSWER TO QUESTION 20 AND RECORD THE NUMBER OF CAREGIVERS WHO KNOW THE CORRECT AGE.	
	188	How many times do you think that a child be vaccinated against yellow fever? RECORD NUMBER OF PARTICIPANTS WHO SAY: • 0 • 1 • 2 • 3 OR MORE.	
	189	Should a child be vaccinated with yellow fever vaccine if the child has: • A mild respiratory illness (for example, cough, runny nose)? • low fever? • diarrhea? • proven recent contact with a sick person? • experienced a previous local reaction from vaccination (redness, etc.)? • child is breastfeeding? • been given another vaccine at the same time? • experienced a previous severe allergic reaction to a vaccine? • severe illness, for example, high fever? • other (describe)? AT THE END OF THIS SECTION, PROVIDE CAREGIVERS WITH CORRECT INFORMATION ON THE ABOVE QUESTIONS; CONTRAINDICATIONS FOR VACCINATION ARE PREVIOUS SEVERE ALLERGIC REACTIONS TO THE VACCINE AND SEVERE ILLNESS.	Number of participants who answered "YES"

	190	FOR EACH QUESTION, WAIT FOR THE RESPONSE AND CONTINUE WITH THE SERIES OF “WHY” AND “WHY NOT” QUESTIONS. PROBE TO SEE IF THESE CONCERNS WOULD PREVENT CAREGIVERS FROM TAKING THEIR CHILDREN TO BE VACCINATED. Do you think yellow fever vaccine is effective in preventing yellow fever disease?	Number of participants who answered “YES”
	191	What are the reasons why caregivers of the community do not take their children to be vaccinated with yellow fever vaccine: (RECORD THE NUMBER OF PARTICIPANTS WHO ANSWERED YES TO THE FOLLOWING ANSWERS.) • concern that yellow fever vaccine may cause serious, life-threatening adverse reactions or death (which ones)? • concern that yellow fever vaccine may cause minor reactions like pain, swelling, redness and fever (which ones)? • concern about buying medicine to prevent or treat the side-effects of the yellow fever vaccine? • concern that my child is already getting too many injections of different vaccines? • other (DESCRIBE)	
SUMMARY	192	REVIEW THE RESPONSES FROM QUESTIONS 185 TO 191. IS THERE A VACCINE MISCONCEPTION IN THE COMMUNITY THAT HAS LED TO CHILDREN NOT BEING VACCINATED?	YES ☐ NO ☐
B.3.3 Obstacles to the vaccination activities (routine and campaigns)			
GD	193	BEFORE STARTING THE DISCUSSION, RECORD THE NUMBER OF PARTICIPANTS IN THE ROOM. Number of participants in the room?	
	194	I am going to read to you a list of possible reasons why caregivers might not take their child to be vaccinated with the yellow fever vaccine during routine vaccination activities and/or mass vaccination campaigns. Please indicate if you think that this is a reason why you do not take your child to be vaccinated: RECORD THE NUMBER OF PARTICIPANTS WHO ANSWERED YES TO THE FOLLOWING ANSWERS. • competing activities at home or outside when the vaccination activities take place? • long queue on the vaccination site? • mother not allowed to leave home? • caregivers will have to pay money to treat any side-effects of the vaccine? • caregivers prefer shaman or traditional medicine? • caregivers have religious objections to vaccination?	Number of persons that answered “YES”

- vaccinator cannot speak or understand the language of the caregivers?
- too costly (consider any payments to the clinic, the cost of getting there, plus the cost of taking time off work)?
- other (DESCRIBE)?

195 I am going to read to you one last list of possible reasons why children in your community might not be vaccinated with the yellow fever vaccine. When you or another caregiver from your community took their child to be vaccinated, did it ever happen that ...:

Number of persons that answered "YES"

RECORD THE NUMBER OF PARTICIPANTS WHO ANSWERED YES TO THE FOLLOWING ANSWERS.

- vaccine was not available at the site?
- vaccinator was not available at the site?
- vaccinator was not friendly?
- vaccinator was not skilled?
- vaccinator did not give the yellow fever vaccine because the child had a mild illness?
- vaccinator did not give the yellow fever vaccine because the child was seriously ill?
- vaccination site location was not suitable?
- timing of vaccination session was not suitable?
- had to pay money for the vaccine or vaccination service?
- other (DESCRIBE)?

SUMMARY 196 REVIEW THE RESPONSES TO QUESTIONS 193 TO 195.

ARE THERE OBSTACLES TO THE VACCINATION ACTIVITIES IN THE COMMUNITY THAT LED TO CHILDREN NOT BEING VACCINATED?

YES ☐ NO ☐

B.3.4 Lack of awareness of yellow fever vaccination campaigns

GD	197	BEFORE STARTING THE DISCUSSION, RECORD THE NUMBER OF PARTICIPANTS IN THE ROOM. Number of participants in the room? Was there one or several yellow fever vaccination campaigns implemented in the district in the last 20 years (REFER TO QUESTION 3)?	Yes ☐	No ☐	If NO, end of the questionnaire
	198	Did you hear about the mass yellow fever vaccination campaign that took place in (YEAR REPORTED IN QUESTION 4)?	Number of persons that answered "YES"		
	199	For the persons who answered "YES", did you hear about the campaign from: - relatives/neighbors? - community leaders? - television, radio, social media, etc.? - health workers at the health centre? - other (DESCRIBE)?	Number of persons that answered "YES"		
SUMMARY	200	REVIEW THE RESPONSES TO QUESTIONS 197 TO 199. IS THERE A LACK OF AWARENESS OF YELLOW FEVER CAMPAIGNS IN THE COMMUNITY THAT MAY HAVE LED TO CHILDREN NOT BEING VACCINATED?	YES ☐	NO ☐	

Comments:



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